

Design and Analysis of Algorithms

Recurrence Relation

National University of Computer and Emerging Sciences,
Islamabad

Recurrences

- The expression:

$$T(n) = \begin{cases} c & n = 1 \\ 2T\left(\frac{n}{2}\right) + cn & n > 1 \end{cases}$$

is a *recurrence*.

- **Recurrence:** an equation that describes a function in terms of its value on smaller functions

Recurrence Examples

$$s(n) = \begin{cases} 0 & n = 0 \\ c + s(n-1) & n > 0 \end{cases}$$

$$s(n) = \begin{cases} 0 & n = 0 \\ n + s(n-1) & n > 0 \end{cases}$$

$$T(n) = \begin{cases} c & n = 1 \\ 2T\left(\frac{n}{2}\right) + c & n > 1 \end{cases}$$

$$T(n) = \begin{cases} c & n = 1 \\ aT\left(\frac{n}{b}\right) + cn & n > 1 \end{cases}$$

Methods to solve recurrence

- **Iteration method**
- **Substitution method**
- **Recursion tree method**
- **Master Theorem**

Solving Recurrences

“iteration method”

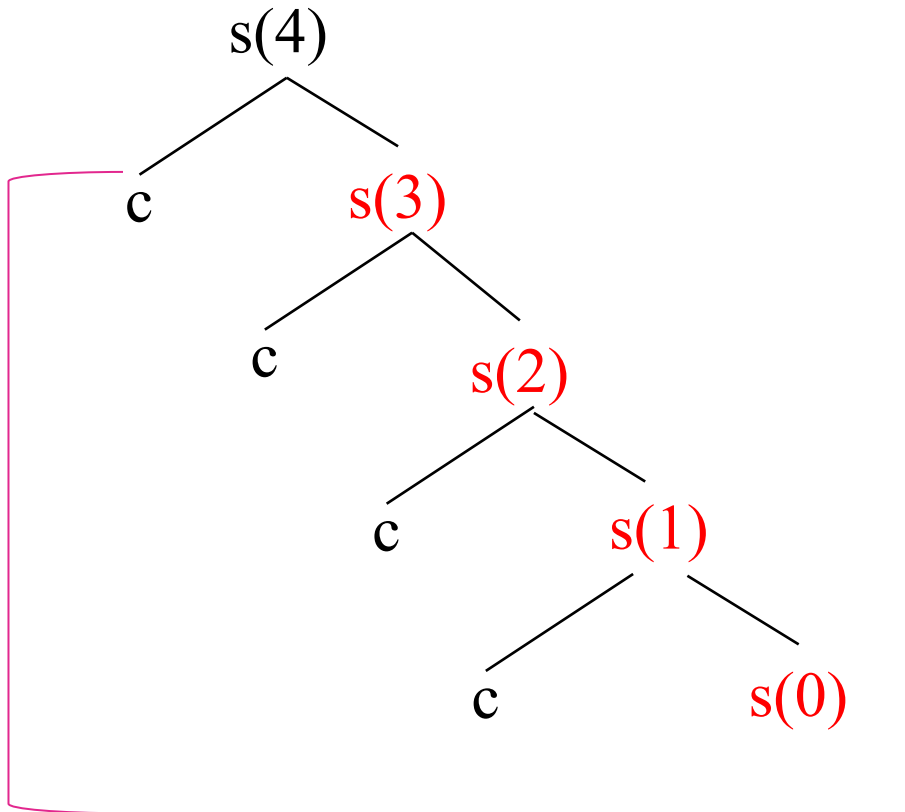
- **Work some algebra to express as a summation**
- **Evaluate the summation**

Example 1

$$s(n) = \begin{cases} 0 & n = 0 \\ c + s(n-1) & n > 0 \end{cases}$$

$$s(n-1) = c + s(n-2)$$

$$s(n-2) = c + s(n-3)$$



$$= 4c + s(0)$$

$$= 4c + 0$$

$$= 4c$$

$$= nc = n$$

$$n=4$$

Example 1

$$s(n) = \begin{cases} 0 & n = 0 \\ c + s(n-1) & n > 0 \end{cases}$$

- $s(4) =$

$$c + s(4-1)$$

$$c + c + s(4-2)$$

$$2c + s(4-2)$$

$$2c + c + s(4-3)$$

$$3c + s(4-3)$$

$$4c + s(4-4)$$

$$= 4c + s(0)$$

$$= 4c + 0 \quad [\text{Stops here}]$$

$$= nc$$

$$s(n-1) = c + s(n-2)$$

$$s(n-2) = c + s(n-3)$$

Example 1

- $s(n) =$
 $c + s(n-1)$

$$s(n) = \begin{cases} 0 & n = 0 \\ c + s(n-1) & n > 0 \end{cases}$$

$$s(n-1) = c + s(n-2)$$

Example 1

- $s(n) =$
 $c + s(n-1)$
 $c + c + s(n-2)$

$$s(n) = \begin{cases} 0 & n = 0 \\ c + s(n-1) & n > 0 \end{cases}$$

$$s(n-1) = c + s(n-2)$$

Example 1

- $s(n) =$
 $c + s(n-1)$
 $c + c + s(n-2)$
 $2c + s(n-2)$

$$s(n) = \begin{cases} 0 & n = 0 \\ c + s(n-1) & n > 0 \end{cases}$$

$$s(n-1) = c + s(n-2)$$

$$s(n-2) = c + s(n-3)$$

Example 1

- $s(n) =$

$c + s(n-1)$

$c + c + s(n-2)$

$2c + s(n-2)$

$2c + c + s(n-3)$

$$s(n) = \begin{cases} 0 & n = 0 \\ c + s(n-1) & n > 0 \end{cases}$$

$$s(n-1) = c + s(n-2)$$

$$s(n-2) = c + s(n-3)$$

Example 1

- $s(n) =$
 $c + s(n-1)$
 $c + c + s(n-2)$
 $2c + s(n-2)$
 $2c + c + s(n-3)$
 $3c + s(n-3)$

$$s(n) = \begin{cases} 0 & n = 0 \\ c + s(n-1) & n > 0 \end{cases}$$

$$s(n-1) = c + s(n-2)$$

$$s(n-2) = c + s(n-3)$$

Example 1

- $s(n) =$

$$c + s(n-1)$$

$$c + c + s(n-2)$$

$$2c + s(n-2)$$

$$2c + c + s(n-3)$$

$$3c + s(n-3)$$

...

$$kc + s(n-k) = ck + s(n-k)$$

$$s(n) = \begin{cases} 0 & n = 0 \\ c + s(n-1) & n > 0 \end{cases}$$

$$s(n-1) = c + s(n-2)$$

$$s(n-2) = c + s(n-3)$$

Example 1

- $s(n) =$

$$c + s(n-1)$$

$$2c + s(n-2)$$

$$3c + s(n-3)$$

...

$$kc + s(n-k) = ck + s(n-k)$$

$$s(n) = \begin{cases} 0 & n = 0 \\ c + s(n-1) & n > 0 \end{cases}$$

$$s(n-1) = c + s(n-2)$$

$$s(n-2) = c + s(n-3)$$

Example 1

$$s(n) = \begin{cases} 0 & n = 0 \\ c + s(n-1) & n > 0 \end{cases}$$

- $s(n) =$

$$c + s(n-1)$$

$$s(n-1) = c + s(n-2)$$

$$2c + s(n-2)$$

$$s(n-2) = c + s(n-3)$$

$$3c + s(n-3)$$

$$4c + s(n-4)$$

...

$$kc + s(n-k) = ck + s(n-k)$$

Example 1

$$s(n) = \begin{cases} 0 & n = 0 \\ c + s(n-1) & n > 0 \end{cases}$$

- So far for $n \geq k$ we have
 - $s(n) = ck + s(n-k)$
- What if $k = n$?
 - $s(n) = cn + s(0) = cn$

Example 1

$$s(n) = \begin{cases} 0 & n = 0 \\ c + s(n-1) & n > 0 \end{cases}$$

- So far for $n \geq k$ we have

- $s(n) = ck + s(n-k)$

- What if $k = n$?

- $s(n) = cn + s(0) = cn$

- So $s(n) = \begin{cases} 0 & n = 0 \\ c + s(n-1) & n > 0 \end{cases}$

- Thus in general

- $s(n) = cn$

Example 2

$$s(n) = \begin{cases} 0 & n = 0 \\ n + s(n-1) & n > 0 \end{cases}$$

- **s(n)**

Example 2

$$s(n) = \begin{cases} 0 & n = 0 \\ n + s(n-1) & n > 0 \end{cases}$$

- $s(n)$
 $= n + s(n-1)$

Example 2

$$s(n) = \begin{cases} 0 & n = 0 \\ n + s(n-1) & n > 0 \end{cases}$$

- $s(n)$
 $= n + s(n-1)$
 $= n + n-1 + s(n-2)$

Example 2

$$s(n) = \begin{cases} 0 & n = 0 \\ n + s(n-1) & n > 0 \end{cases}$$

- $s(n)$
 $= n + s(n-1)$
 $= n + n-1 + s(n-2)$
 $= n + n-1 + n-2 + s(n-3)$

Example 2

$$s(n) = \begin{cases} 0 & n = 0 \\ n + s(n-1) & n > 0 \end{cases}$$

- $s(n)$
 $= n + s(n-1)$
 $= n + n-1 + s(n-2)$
 $= n + n-1 + n-2 + s(n-3)$
 $= n + n-1 + n-2 + n-3 + s(n-4)$

Example 2

$$s(n) = \begin{cases} 0 & n = 0 \\ n + s(n-1) & n > 0 \end{cases}$$

- $s(n)$
 $= n + s(n-1)$
 $= n + n-1 + s(n-2)$
 $= n + n-1 + n-2 + s(n-3)$
 $= n + n-1 + n-2 + n-3 + s(n-4)$
 $= \dots$
 $= n + n-1 + n-2 + n-3 + \dots + n-(k-1) + s(n-k)$

Example 2

$$s(n) = \begin{cases} 0 & n = 0 \\ n + s(n-1) & n > 0 \end{cases}$$

- $s(5)$

$$= 5 + s(5-1)$$

$$= 5 + S(4)$$

$$= 5 + 4 + s(5-2)$$

$$= 5 + 4 + S(3)$$

$$= 5 + 4 + 3 + s(5-3)$$

$$= 5 + 4 + 3 + S(2)$$

$$= 5 + 4 + 3 + 2 + s(5-4)$$

$$= 5 + 4 + 3 + 2 + S(1)$$

$$= 5 + 4 + 3 + 2 + 1 + s(5-5)$$

$$= 5 + 4 + 3 + 2 + 1 + S(0)$$

Example 2

$$s(n) = \begin{cases} 0 & n = 0 \\ n + s(n-1) & n > 0 \end{cases}$$

- $s(n)$
 $= n + s(n-1)$
 $= n + n-1 + s(n-2)$
 $= n + n-1 + n-2 + s(n-3)$
 $= n + n-1 + n-2 + n-3 + s(n-4)$
 $= \dots$
 $= n + n-1 + n-2 + n-3 + \dots + n-(k-1) + s(n-k)$

Example 2

$$s(n) = \begin{cases} 0 & n = 0 \\ n + s(n-1) & n > 0 \end{cases}$$

- $s(n)$

$$= n + s(n-1)$$

$$= n + n-1 + s(n-2)$$

$$= n + n-1 + n-2 + s(n-3)$$

$$= n + n-1 + n-2 + n-3 + s(n-4)$$

$$= \dots$$

$$= n + n-1 + n-2 + n-3 + \dots + n-(k-1) + s(n-k)$$

$$=$$

$$\sum_{i=n-k+1}^n i + s(n-k)$$

Example 2

$$s(n) = \begin{cases} 0 & n = 0 \\ n + s(n-1) & n > 0 \end{cases}$$

- So far for $n \geq k$ we have

$$\sum_{i=n-k+1}^n i + s(n-k)$$

Example 2

$$s(n) = \begin{cases} 0 & n = 0 \\ n + s(n-1) & n > 0 \end{cases}$$

- So far for $n \geq k$ we have

$$\sum_{i=n-k+1}^n i + s(n-k)$$

- What if $k = n$?

$$\begin{aligned} S(5) &= 5 + 4 + 3 + 2 + 1 + s(5-5) \\ &= 5 + 4 + 3 + 2 + 1 + S(0) \end{aligned}$$

Example 2

$$s(n) = \begin{cases} 0 & n = 0 \\ n + s(n-1) & n > 0 \end{cases}$$

- So far for $n \geq k$ we have

$$\sum_{i=n-k+1}^n i + s(n-k)$$

- What if $k = n$?

$$\sum_{i=1}^n i + s(0) = \sum_{i=1}^n i + 0 = n \frac{n+1}{2}$$

Example 2

$$s(n) = \begin{cases} 0 & n = 0 \\ n + s(n-1) & n > 0 \end{cases}$$

- So far for $n \geq k$ we have

$$\sum_{i=n-k+1}^n i + s(n-k)$$

- What if $k = n$?

$$\sum_{i=1}^n i + s(0) = \sum_{i=1}^n i + 0 = n \frac{n+1}{2}$$

- Thus in general

$$s(n) = n \frac{n+1}{2}$$

Divide and Conquer

The divide-and-conquer paradigm

What is a paradigm? One that serves as a pattern or model.

- **Divide** the problem into a number of subproblems
- **Conquer** the subproblems by solving them recursively. If small enough, just solve directly without recursion
- **Combine** the solutions of the subproblems into the solution for the original problem

Analyzing Divide-and-Conquer Algorithms

Let $T(n)$ be the running time on a problem of size n .

If problem is small enough, then solution takes constant time (this is a boundary condition, which will be assumed for all analyses)

It takes time to divide the problem into sub-problems at the beginning. Denote this work by $D(n)$.

Analyzing Divide-and-Conquer Algorithms

It takes time to combine the solutions at the end. Denote this by $C(n)$.

If we divide the problem into ‘a’ problems, each of which is ‘1/b’ times the original size (n), and since the time to solve a problem with input size ‘n/b’ is $T(n/b)$, and this is done ‘a’ times, the total time is:

$$T(n) = \Theta(1), n \text{ small (or } n \leq c \text{)}$$

$$T(n) = aT(n/b) + D(n) + C(n)$$

Example

Function(int number)

if number<=1

then return;

else

Function(number/2)

Function(number/2)

for(i=1 to number)

Display i;

It follows that

$$T(n) = 2 T(n/2) + \textcircled{cn}$$

number of sub-problems

Sub-problem size

work done dividing and combining